

TrES-2b

R-2495

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_120903 · created 2026-08-24T01:45:01+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456173.77542 +/- 0.00017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.038
Transit depth (Rp/Rs)^2	2.22 +/- 1.14 [%]
Orbital Inclination (inc)	84.27 +/- 2.33
Airmass coefficient 1 (a1)	1.039 +/- 0.025
Airmass coefficient 2 (a2)	-0.026 +/- 0.023
Scatter in the residuals of the lightcurve fit is	2.4 %
Best Comparison Star	#2 - [293, 196]
Optimal Aperture	3.41
Optimal Annulus	14.1
Transit Duration (day)	0.072 +/- 0.035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.149 $\pm$ 0.038 vs 0.1254 (deviation 0.62 σ)
Duration fit vs published	0.072 d vs 0.0746 d (deviation 0.07 σ)
Mid-time O-C	-0.4 min
Depth SNR	3.9
Residual scatter	2.4 %

Light curve

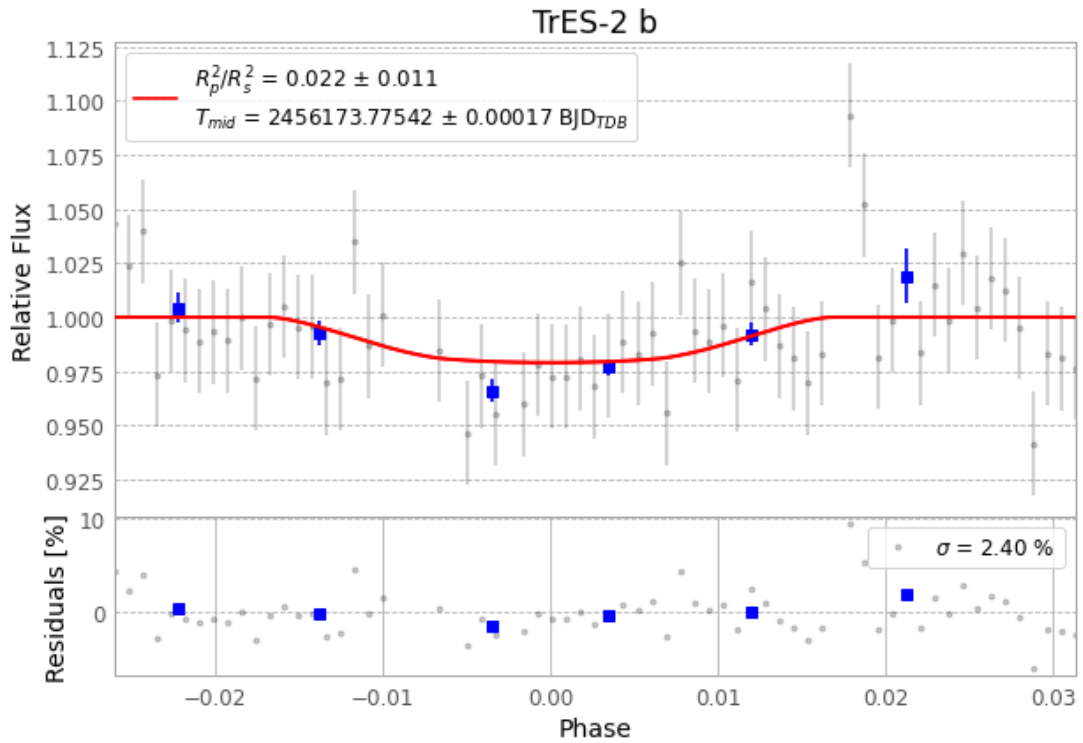


Figure 1 — FinalLightCurve_TrES-2 b_2012-09-02.png (SHA-256 99c7ba8cf2fed5c9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [287, 261], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 73e6d1c06f2ae693...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

69 FITS frames (45 MB), TRES-2120903050012.FITS → TRES-2120903082412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-120903.log (c49d1076427a...), FinalLightCurve_TrES-2 b_2012-09-02.png (99c7ba8cf2fe...), FinalLightCurve_TrES-2 b_2012-09-02.csv (42f8cba98a16...)

Compute resources

Wall time 143.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-24 01:45Z.